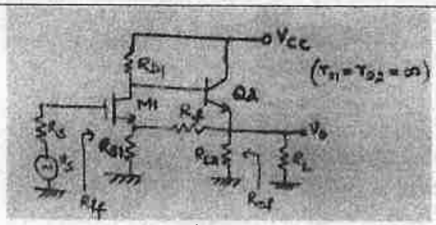
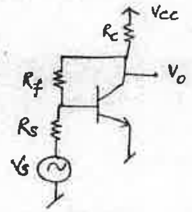
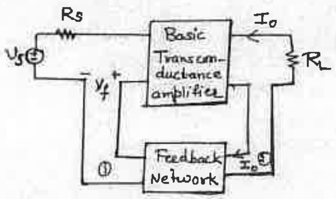
	 Fig. 3(b)		
3(c)	For the circuit shown in Fig. 3(c), with following parameters: $R_C = 4k\Omega$, $R_F = 40k\Omega$, $R_S = 10k\Omega$, $r_\pi = 1.1k\Omega$, $\beta = 50$, find: (i) β (ii) A (iii) A_f (iv) R_{if}, and (v) R_{of}.  Fig. 3(c)	(1+1+0.5 +1+0.5)	CO3
Q4	Attempt any 2 parts of the following		
4(a)	(i) Draw the circuit of MOSFET based Wilson current mirror. (ii) Derive the expression for its output resistance. (iii) List ideal characteristics of a current mirror.	(1+2+1)	CO4
4(b)	i) Draw the circuit of class B push-pull amplifier and derive its relation of conversion (overall) efficiency. ii) A class A, common emitter amplifier operates from $V_{CC} = 20V$, draws a no signal current of 5A and feeds a load of 40Ω through a step-up transformer of $n_2/n_1 = 3.16$. Find: (a) Maximum ac signal power output. (b) maximum dc power input (c) conversion efficiency at maximum signal input.	(2+2)	CO4
4(c)	i) The feedback circuit representable by Fig. 4(c) uses ideal trans-conductance amplifier and operates with $V_S = 100mV$, $V_f = 95mV$ and $I_o = 10mA$. Derive the expression for A_f and determine the corresponding values of A and β. Include the unit in each answer.  Fig. 4(c) ii) Find R_{if} and R_{of} in terms of R_i, A and β for the topology shown in Fig. 4(c).	(2+2)	CO3
Q5	Attempt any 2 parts of the following		
5(a)	(i) State the Barkhausen Criteria. (ii) Draw Opamp based RC phase shift oscillator (iii) Derive its expressions for condition of sustained oscillation and frequency of oscillation.	(1+2+1)	CO5
5(b)	(i) Draw the circuit diagram of an op-amp based triangular waveform generator. (ii) Explain the principle of operation and derive the expression for frequency of the generated waveform in terms of circuit parameters.	(2+2)	CO5
5(c)	(i) With the help of a neat diagram, explain the internal structure of the 555 Timer IC and describe its working in astable multivibrator mode. (ii) Derive the expressions for frequency and duty cycle of the output waveform.	(3+1)	CO5

Total No. of Page(s): 03.. Roll No.................

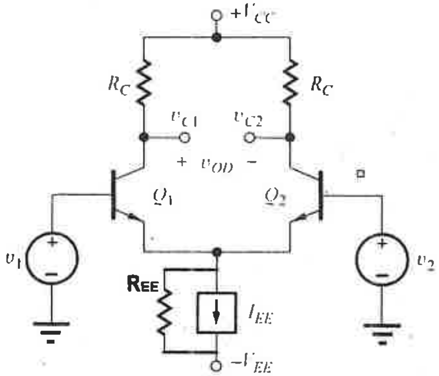
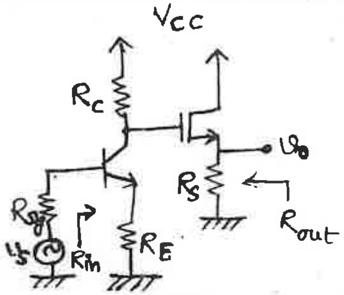
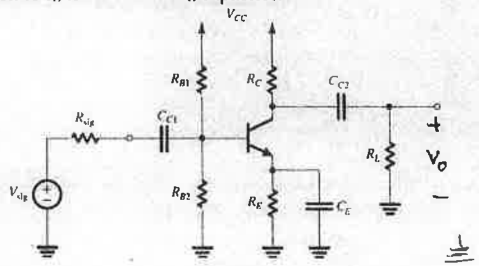
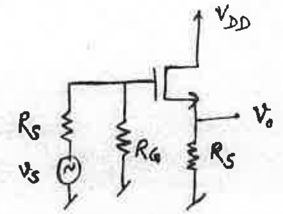
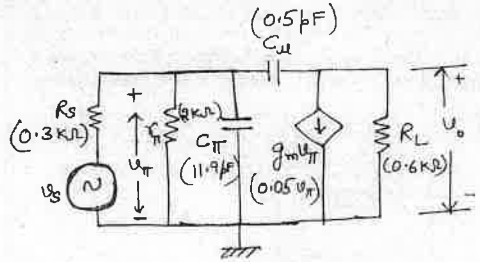
THIRD SEMESTER **B.Tech. (ECE)**

END SEM. EXAMINATION, Nov.-Dec. 2025

Course Code: ECECC07/EIECC07 (Backlog/Old Scheme)
Course Title: Microelectronics Circuits and Applications

Time: 03:00 Hrs. Max. Marks: 40

Note: Attempt all the five questions. All questions carry equal marks. Missing data/information if any will be suitably assumed and mentioned in the answer.

Q. No.	Questions	Marks	CO
Q1	Attempt any 2 parts of the following		
1(a)	i) What is the Q-point for the transistors in the amplifier (I_C and V_{CQ}) in Fig. 1(a), if V_{CC} = 12 V, V_{EE} = 12 V, I_{EE} = 400μA, β = 100, R_{EE} = 270 kΩ, R_C =47 kΩ, and V_A=∞. ii) Calculate for circuit in Fig. 1(a): differential-mode gain, common-mode gain, CMRR, and differential-mode input resistance?  Fig. 1(a)	(1+3)	CO1
1(b)	For the amplifier circuit shown in Fig. 1(b): i) Draw small signal equivalent circuit. ii) Find overall gain (Av) iii) Input resistance (R_{in}) and iv) Output resistance (R_{out}).  Fig. 1(b)	(1+1+1+1)	CO1
1(c)	i) Draw the circuit of MOSFET based differential amplifier with current mirror active load with single ended output. ii) Derive its output resistance and iii) Differential Gain. iv) Determine the advantage of this differential amplifier over single ended differential amplifier without current mirror load.	(1+1+1+1)	CO1
Q2	Attempt any 2 parts of the following		
2(a)	For the common emitter amplifier shown in Fig. 2(a), find the value of C_{C1}, C_{C2} and C_E for f_L = 100 Hz, considering R_{B1} = R_{B2} = 200kΩ, R_C = 8kΩ, R_L = 5kΩ, R_{sig} = 5kΩ, R_E = 5kΩ, β = 100, g_m = 40mA/V, and r_π = 2.5kΩ. For the calculations, assume that 80% of the contribution to f_L is due to the C_E capacitor.  Fig. 2(a)	(4)	CO2
2(b)	For the circuit shown in Fig. 2(b) draw small signal high frequency equivalent circuit and find higher cut off frequency (f_H) by assuming the dominant pole approximation is valid.  Fig. 2(b)	(4)	CO2
2(c)	For the common emitter amplifier whose high-frequency equivalent circuit is shown in Fig. 2(c), determine coefficients (i) a_1 and (ii) a_2.  Fig. 2(c)	(2+2)	CO2
Q 3	Attempt any 2 parts of the following		
3(a)	(i) Draw the shunt-series feedback amplifier topology in signal flow graph form/block diagram and find gain with feedback (A_f), assuming that negative feedback is applied. (ii) Find its R_{if} and R_{of} in terms of A and β.	(2+2)	CO3
3(b)	For the circuit shown in Fig. 3(b) determine the topology of feedback and find: (i) β circuit (ii) Draw modified-A circuit, and obtain (iii) input resistance R_{if}, (iv) output resistance R_{of}, and (v) closed-loop voltage gain A_f.	(1+0.5+1 +1+0.5)	CO3